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Operationalizing Verification and Validation for Text Corpora: A Parameter-Free Information-Theoretic Framework

A presentation in partial fulfilment for the award of Masters (MSc.) Computing for Data Science

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THE PROBLEM

The corpus is the least-audited part of an NLP the pipeline

Empirical NLP rests on corpora; yet the corpus is rarely subjected to the quality control engineering demands of any other component. We lack a systematic verification & validation (V&V) process for corpora.

A 2022 manual audit of 205 web-crawled multilingual corpora found systematic quality failures concentrated in lower-resource languages: at least 15 corpora contained no usable text at all, and 82 were mislabelled or carried nonstandard language codes (Kreutzer 2022).

> *a void at the heart of corpus linguistics*

The defects that break corpora are exactly the ones a routine V&V process would catch - if we had one.

Where does a defect leave its evidence?

Our framework rests on one principle: different corpus defects leave their evidence in different places. Find the place, and you know how to look.

Structural defect

Evidence lives

BETWEEN documents

A duplicate, a truncated stub, a repeated boilerplate line - each is a relationship among documents. The aggregate word distribution barely moves.

Content defect

Evidence lives

IN the aggregate

A foreign-language passage or a corrupted encoding changes the distribution of characters itself. The whole corpus's profile shifts.

Evidence location dictates the measurement level

If the principle is right, it makes a testable prediction: each defect is caught by the instrument that looks where its evidence lives and, missed by the other.

Defect	Evidence lives in	Instrument
Duplication	between documents	**Document-level**
Truncation	document lengths	**Document-level**
Boilerplate	repeated spans	**Document-level**
Foreign text	aggregate content	**Corpus-level**
Encoding corruption	aggregate content	**Corpus-level**

RGT: a parameter-free instrument for content defects

The prediction needs a corpus-level instrument to test it. Comparing two corpora by information divergence needs a probability for every n-gram, including unseen ones. Standard add-k smoothing forces an arbitrary parameter. Reciprocal Good-Turing removes it.

1

Mass from the data

Reserve the Good-Turing singleton estimate, $\theta = n1 / N$. No parameter, a property of the corpus.

2

Shape from the partner

Spread that mass over unseen n-grams like the OTHER corpus's distribution, not uniformly. Where we lack evidence, assume the corpus resembles its comparison partner.

Why it fits the role

- No free parameter
- Self-adapts across n-gram orders & languages
- Conservative: keeps signal, suppresses noise

θ self-adapts: 0.005 to 0.18 across our corpora; ~100x smaller for character- than word-n-grams. No fixed k was able to do this.

Controlled injection supplies ground truth

To TEST whether an instrument detects a defect, we need the right answer. We manufacture it: start clean, inject a known defect at a known rate.

1

Clean corpus

split into reference + test halves

>

2

Inject

known defect at rate r

>

3

Measure

divergence: reference vs test

>

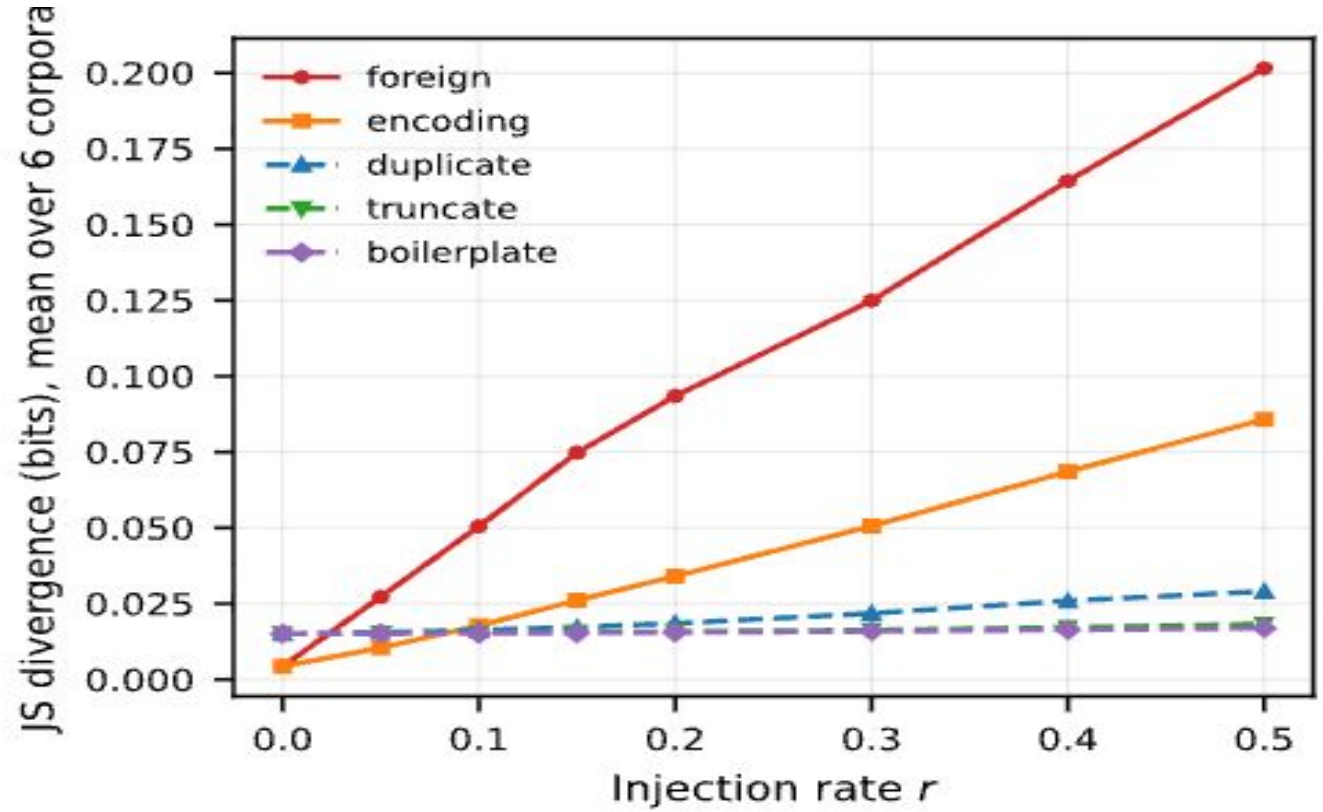
4

Known answer

detection is now falsifiable

Five defect classes - eight rates - six corpora - ten seeds = 2,400 corpus-level + 360 document-level measurements

Does the instrument track the defect?



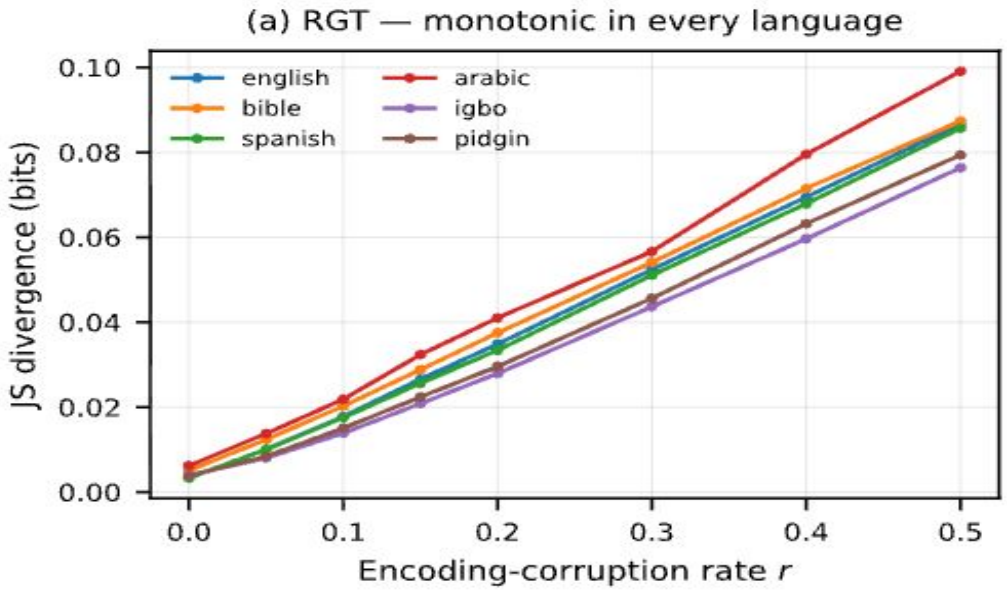
The prediction, visible

- Content defects (red, gold) climb steeply; the corpus-level instrument sees them.
- Structural defects stay flat; corpus-level cannot see them, as predicted.
- **The flatness is the prediction coming true.**

The standard baseline moves the WRONG way

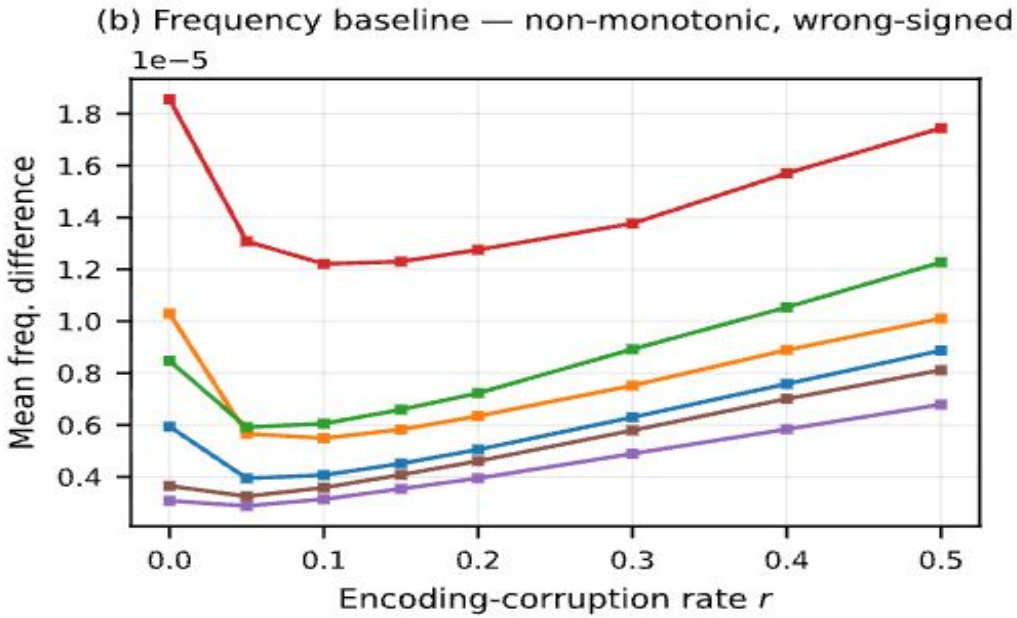
Under light encoding corruption, the common frequency-difference baseline reports a MORE corrupted corpus as MORE similar to clean in every one of the six languages. RGT does not.

Our instrument (RGT)



rises everywhere

Frequency baseline



dips first - the wrong direction

Right instrument, every language

Structural defects - CORPUS level

d <= 0.8

Nearly invisible - the wrong instrument, exactly as predicted.

Same defects - DOCUMENT level

AUC = 1.000

Perfect detection, zero variance - all six languages, every rate. 36 / 36 per-corpus scores.

Move a defect to the instrument that looks where its evidence lives: detection jumps from near-zero to perfect. The principle predicted this, language-independently.

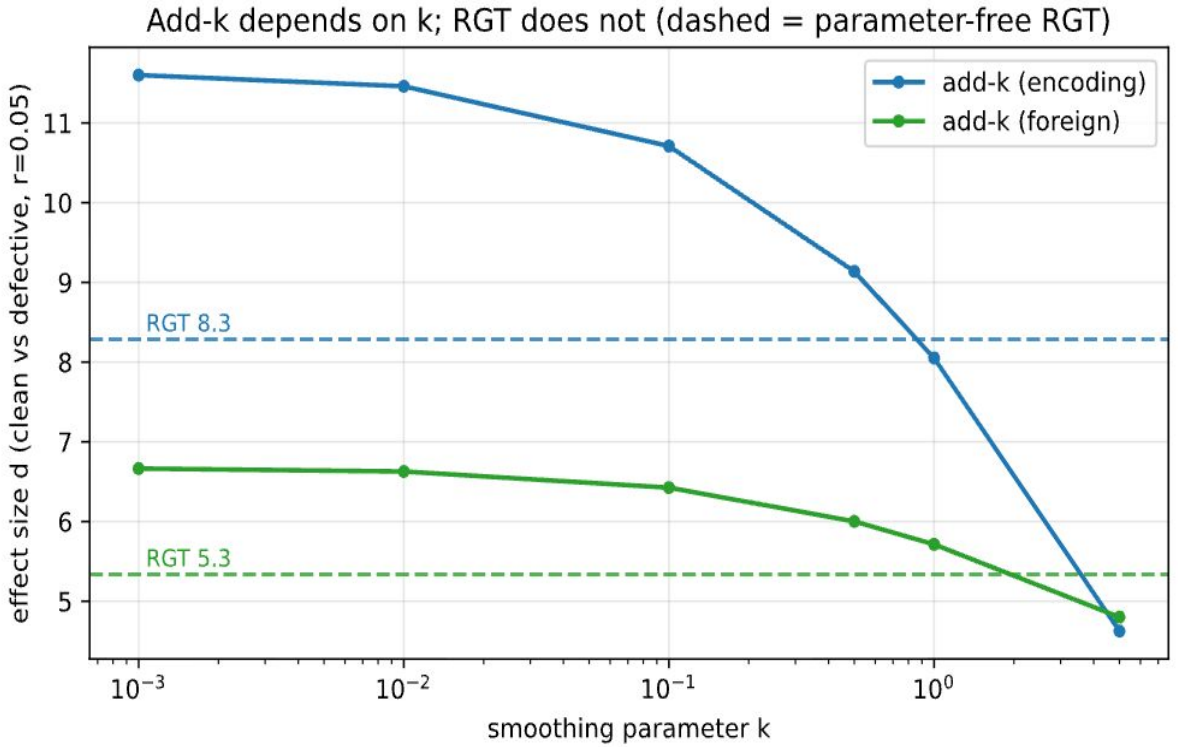
It survives the hard case, with no knob to turn

Same-script contamination

Pidgin <- English - no script change

d = 3.5

Still detected strongly at 5% contamination. The instrument does not rely on a change of script.



RGT (encoding) = 8.3 - no parameter to choose

THE PIPELINE, IN ONE VIEW

Corpus quality, made measurable.

1 Principle

Defects leave evidence in different places - between documents, or in the aggregate.

2 Prediction

Evidence location dictates the measurement level. Falsifiable.

3 Instrument + proof

RGT (parameter-free) at the corpus level; controlled injection supplies ground truth; the prediction holds in 5 languages.

Reproducible: fixed seeds, released code & data. Next: scripts that need segmentation - Chinese, Japanese, Thai.

Diolch!